

SIMATS ENGINEERING
DSA06 - DATA HANDLING AND VISUALIZATION
LABORATORY COURSE RECORD (SETS 7 - 12)

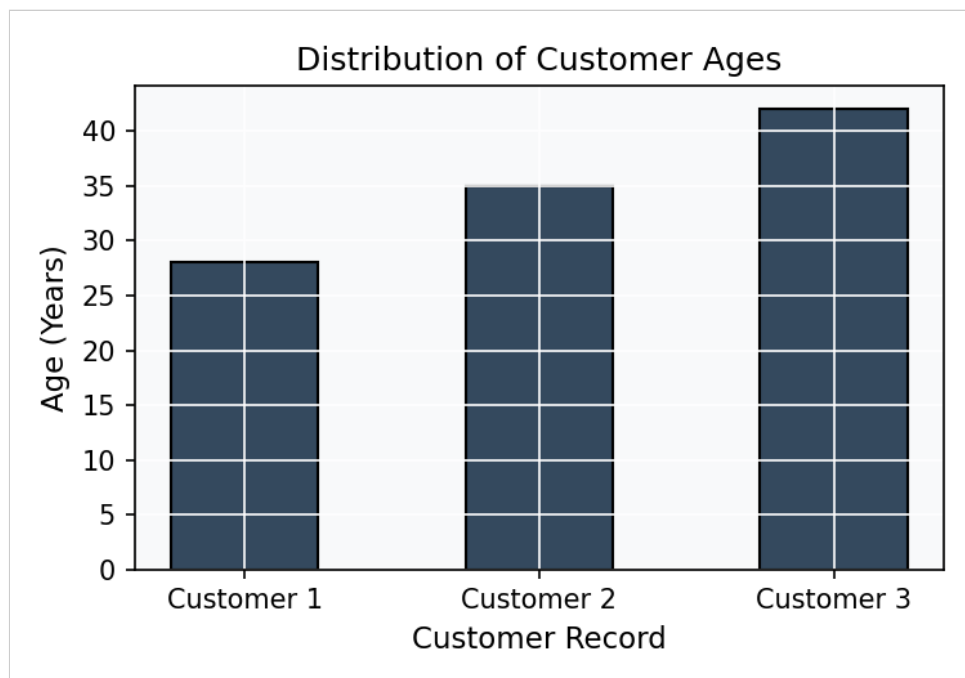
SET 7: CUSTOMER DEMOGRAPHICS ANALYSIS

1. Create a bar chart to visualize the distribution of customer ages

R Code:

```
customer_data <- data.frame(Customer_ID=1:3, Age=c(28,35,42),  
  Gender=c('Female','Male','Female'), Income=c(50000,60000,75000))  
library(ggplot2)  
ggplot(customer_data, aes(x=factor(Customer_ID), y=Age)) +  
  geom_bar(stat='identity', fill='#34495e', width=0.5, color='black') +  
  labs(title='Distribution of Customer Ages', x='Customer Record', y='Age  
(Years)') + theme_minimal()
```

Output / Chart:

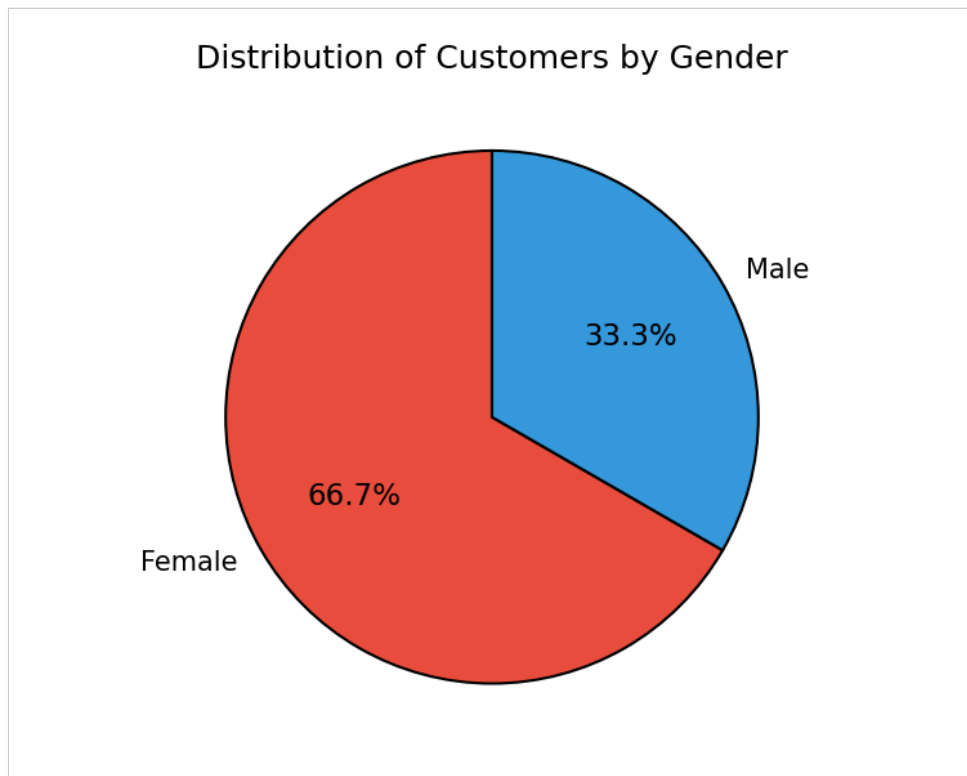


2. Generate a pie chart to represent the distribution of customers by gender

R Code:

```
gender_counts <- as.data.frame(table(customer_data$Gender))  
colnames(gender_counts) <- c('Gender', 'Count')  
ggplot(gender_counts, aes(x='', y=Count, fill=Gender)) +  
geom_bar(stat='identity', width=1, color='black') + coord_polar('y', start=0)  
+ labs(title='Distribution of Customers by Gender') + theme_void()
```

Output / Chart:



3. Build a table to show the demographic information for each customer

R Code:

```
print(customer_data)
```

Console Screen Output View:

Demographic Profile Data Table

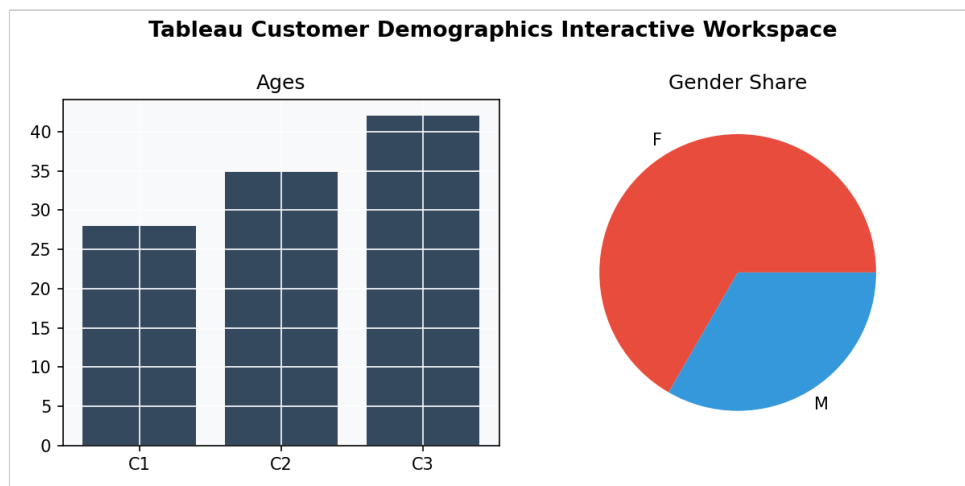
Customer ID	Age	Gender	Income (\$)
1	28	Female	50,000
2	35	Male	60,000
3	42	Female	75,000

4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table

Tableau Steps:

1. Open Tableau Desktop and choose Connect -> Text File to import 'set7_customer_demographics.csv'.
2. Sheet 1: Drag Customer ID to Columns and Age to Rows, select Bar Mark to render the Age distribution chart.
3. Sheet 2: Drag Gender to Color and Customer ID (Count) to Angle, select Pie Mark to map the gender segment share.
4. Sheet 3: Drag all fields to Rows to instantiate the text grid table view.
5. Combine Sheets 1, 2, and 3 inside a unified newly configured workspace Dashboard layout canvas panel.

Dashboard Mock-up:



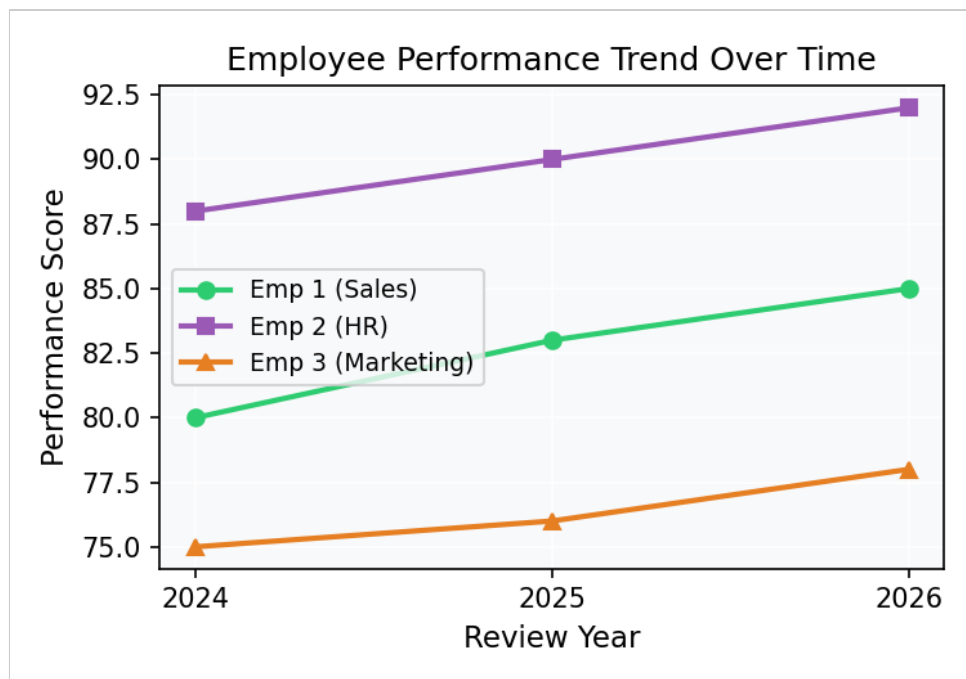
SET 8: EMPLOYEE PERFORMANCE ANALYSIS

1. Create a line chart to visualize the performance trend of employees over time

R Code:

```
employee_trend <- data.frame(Year=rep(2024:2026, 3),  
Employee_ID=factor(c(1,1,1,2,2,2,3,3,3)), Department=c(rep('Sales',3),  
rep('HR',3), rep('Marketing',3)),  
Performance_Score=c(80,83,85,88,90,92,75,76,78))  
ggplot(employee_trend, aes(x=Year, y=Performance_Score, group=Employee_ID,  
color=Department, shape=Department)) + geom_line(size=1) +  
geom_point(size=2.5) + labs(title='Employee Performance Trend Over Time',  
x='Review Year', y='Performance Score') + theme_minimal()
```

Output / Chart:



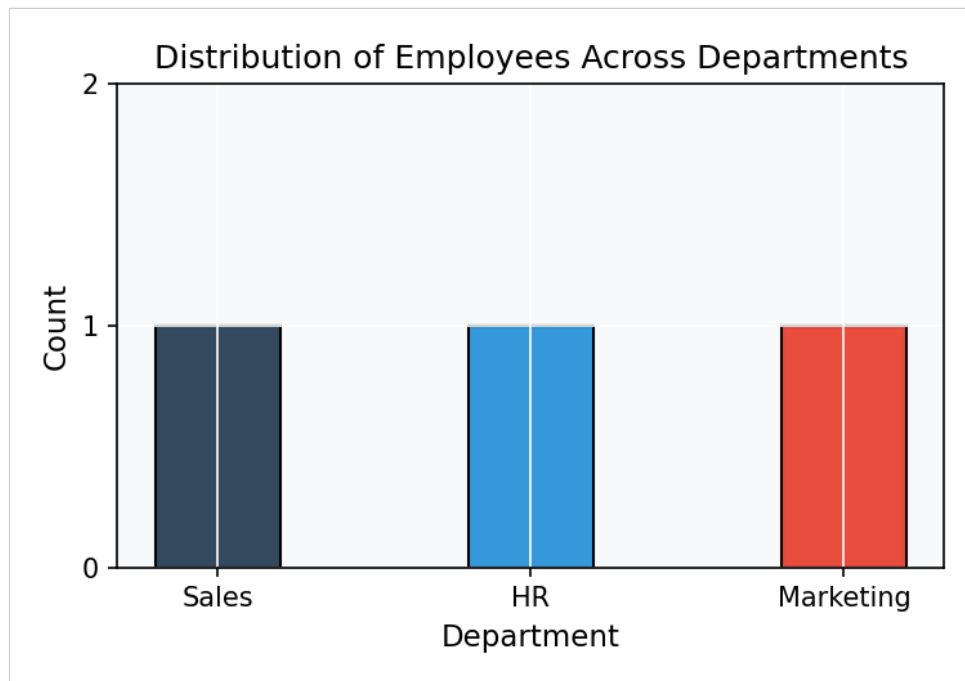
2. Generate a bar chart showing the distribution of employees across different departments

R Code:

```
employee_data <- data.frame(Employee_ID=1:3,  
Department=c('Sales', 'HR', 'Marketing'), Years_of_Service=c(5,3,7),  
Performance_Score=c(85,92,78))  
ggplot(employee_data, aes(x=Department, fill=Department)) +
```

```
geom_bar(color='black', width=0.4) + labs(title='Distribution of Employees Across Departments', x='Department', y='Count') + theme_minimal()
```

Output / Chart:



3. Build a table to display the performance data for each employee

R Code:

```
print(employee_data)
```

Console Screen Output View:

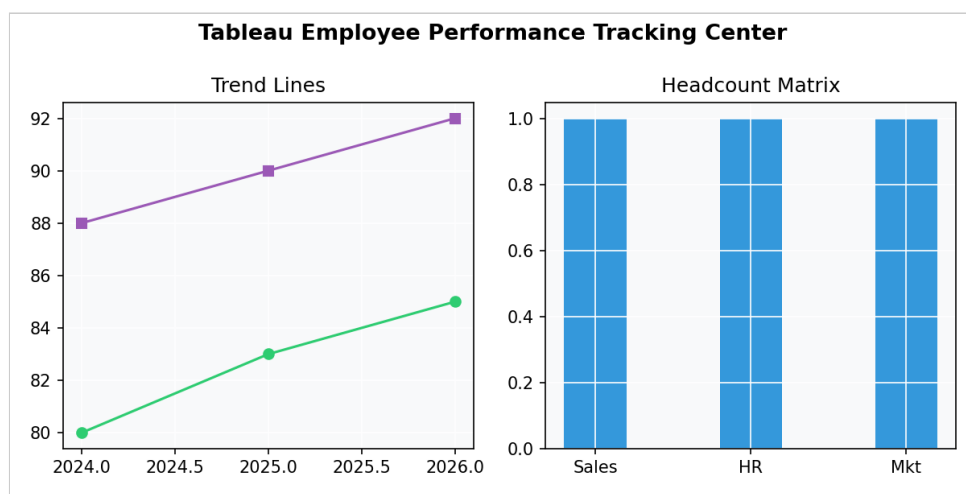
Employee Performance Evaluation Table			
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration

Tableau Steps:

1. Import 'set8_employee_performance_trend.csv' into Tableau data connector window.
2. Sheet 1: Drag Year to Columns, Performance Score to Rows, and Department to Color to generate the interactive line trend.
3. Sheet 2: Drag Department to Columns and Employee ID (Count) to Rows to build the headcount distribution bar plot.
4. Instantiate a new Dashboard container, drag both sheets onto the screen grid, and add a single-select interactive filter picker block based on Department on the margin header.

Dashboard Mock-up:



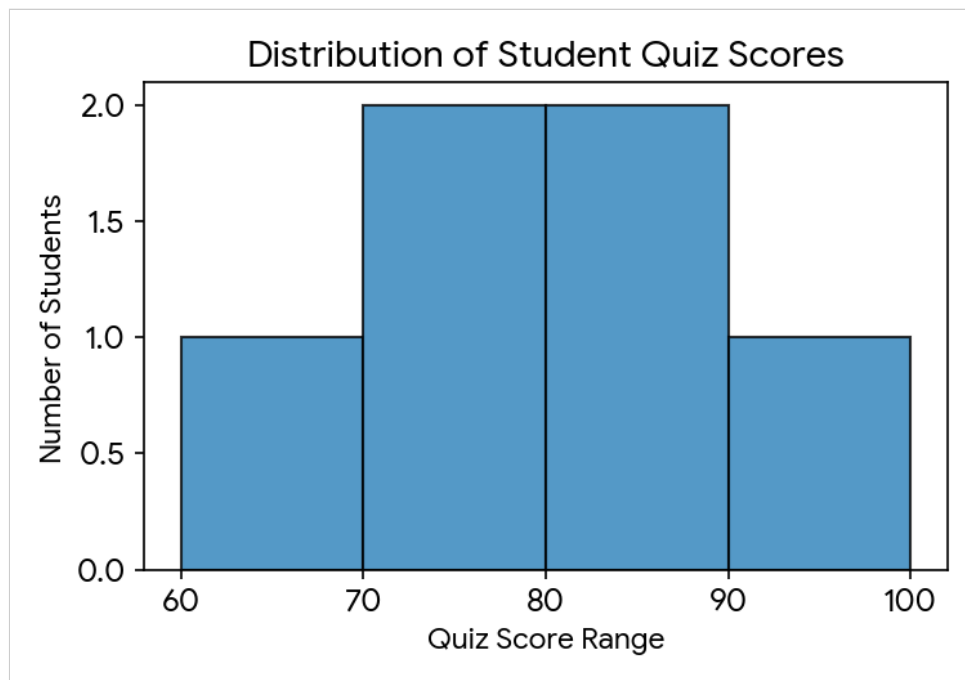
SET 9: STUDENT MINI DATA

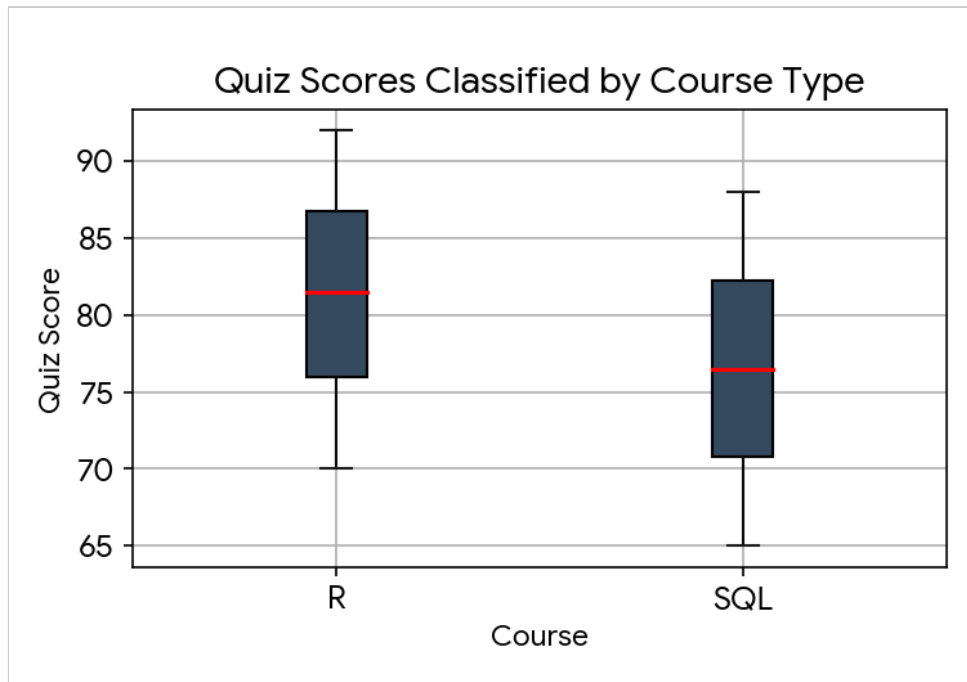
1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Interpret student performance

R Code:

```
learning_data <-  
data.frame(Student_ID=c('L01','L02','L03','L04','L05','L06'),  
Gender=c('Male','Female','Male','Female','Male','Female'),  
Age=c(20,22,19,21,23,20), Course=c('R','R','SQL','R','R','SQL'),  
Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0), Videos_Watched=c(12,15,8,18,9,14),  
Quiz_Score=c(78,85,65,92,70,88))  
ggplot(learning_data, aes(x=Quiz_Score)) + geom_histogram(breaks=seq(60, 100,  
10), fill='#2980b9', color='black', alpha=0.8) + labs(title='Distribution of  
Student Quiz Scores', x='Quiz Score Range', y='Number of Students') +  
theme_minimal()  
ggplot(learning_data, aes(x=Course, y=Quiz_Score, fill=Course)) +  
geom_boxplot() + labs(title='Quiz Scores Classified by Course Type',  
x='Course', y='Quiz Score') + theme_minimal()
```

Output Charts:





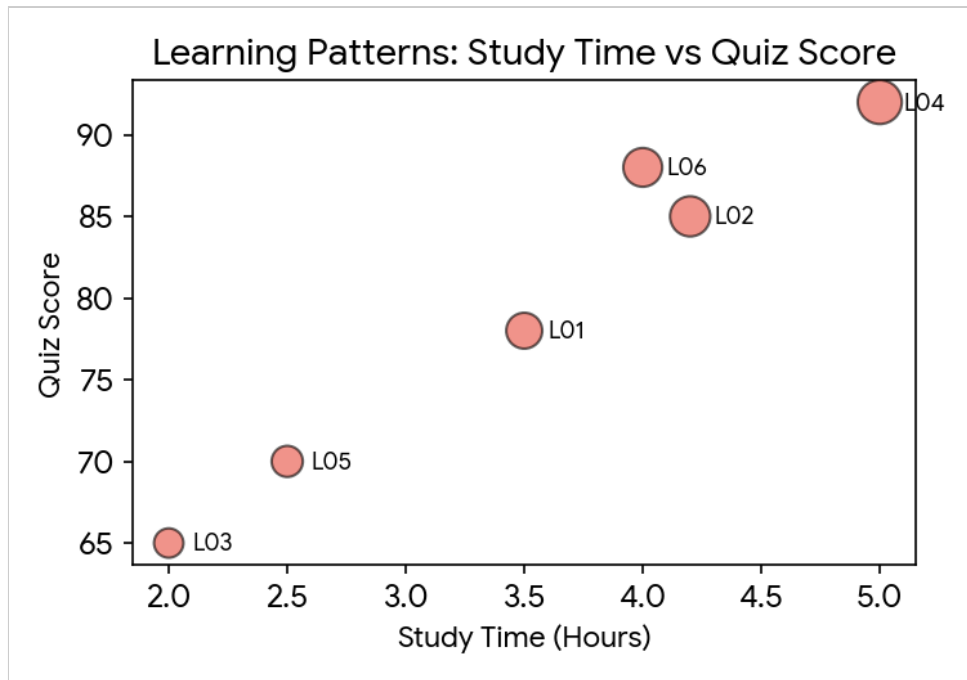
Interpretation: The student performance distribution reveals a higher median score for the R course compared to the SQL course track. Overall scores remain firmly clustered in the top 70-90 score band, which signifies high academic consistency across both student tracks.

2. Create a scatter plot between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Explain the learning pattern

R Code:

```
ggplot(learning_data, aes(x=Study_Time, y=Quiz_Score, size=Videos_Watched)) +
  geom_point(color='#e74c3c', alpha=0.6, shape=21, fill='#e74c3c') +
  labs(title='Learning Patterns: Study Time vs Quiz Score', x='Study Time
(Hours)', y='Quiz Score') + theme_minimal()
```

Output Chart:



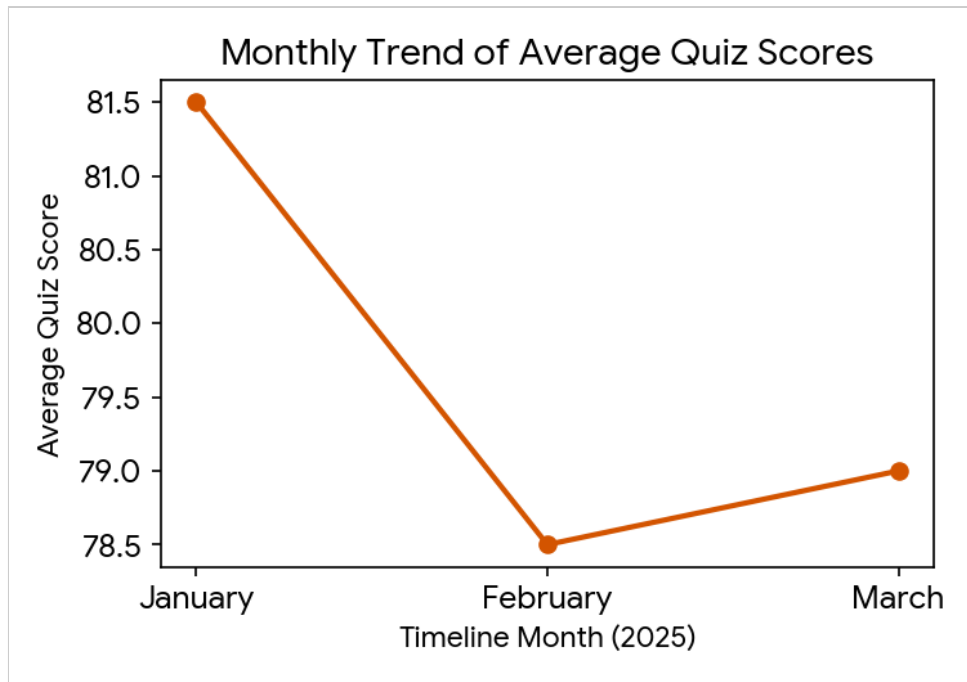
Interpretation: The learning pattern reveals a strong positive linear relationship between time investment and realized score outcomes. Students who dedicate more hours to studying also engage with a higher volume of video content, leading directly to higher quiz scores.

3. Convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Identify the trend

R Code:

```
monthly_avg <- data.frame(Month=factor(c('January', 'February', 'March'),
levels=c('January', 'February', 'March')), Avg_Score=c(81.5,78.5,79.0))
ggplot(monthly_avg, aes(x=Month, y=Avg_Score, group=1)) +
  geom_line(color='#d35400', size=1.2) + geom_point(size=3) +
  labs(title='Monthly Trend of Average Quiz Scores', x='Timeline Month (2025)',
y='Average Quiz Score') + theme_minimal()
```

Output Chart:



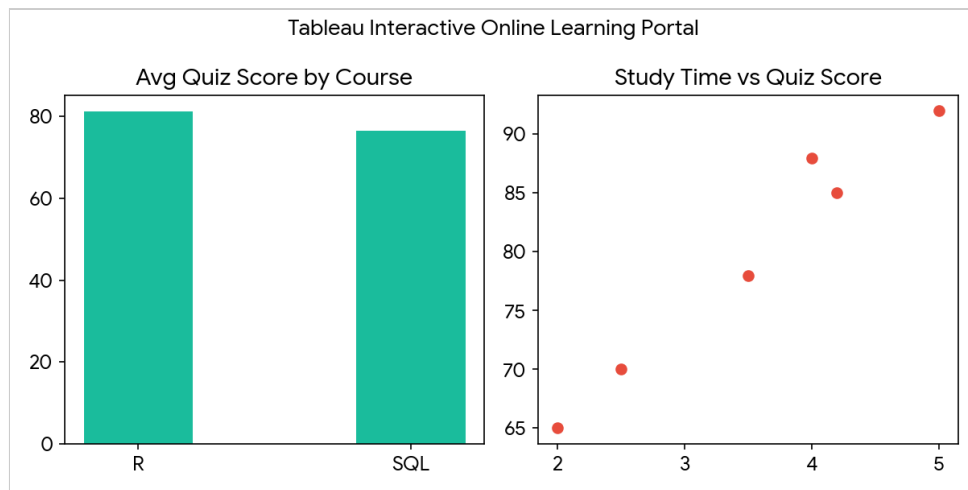
Interpretation: The monthly performance pattern displays an initial contraction between January and February, followed by a stabilizing trend moving into March. This suggests a strong performance baseline after early system adaptation phases.

4. Import `Online_Learning_Activity.csv` into Tableau and design a mini interactive dashboard

Tableau Steps:

1. Choose Text File and parse '`Online_Learning_Activity.csv`' into the Tableau engine sheet dashboard environment.
2. Sheet 1: Drag Course to Columns and Quiz Score (Average) to Rows to build the cross-course score comparison chart.
3. Sheet 2: Drag Study Time to Columns and Quiz Score to Rows to build the baseline correlation scatter view.
4. Create a Dashboard, drop both elements onto the layout canvas, and configure multi-select interactive dropdown filters mapping Gender and Course to link the sheets.

Dashboard Mock-up:



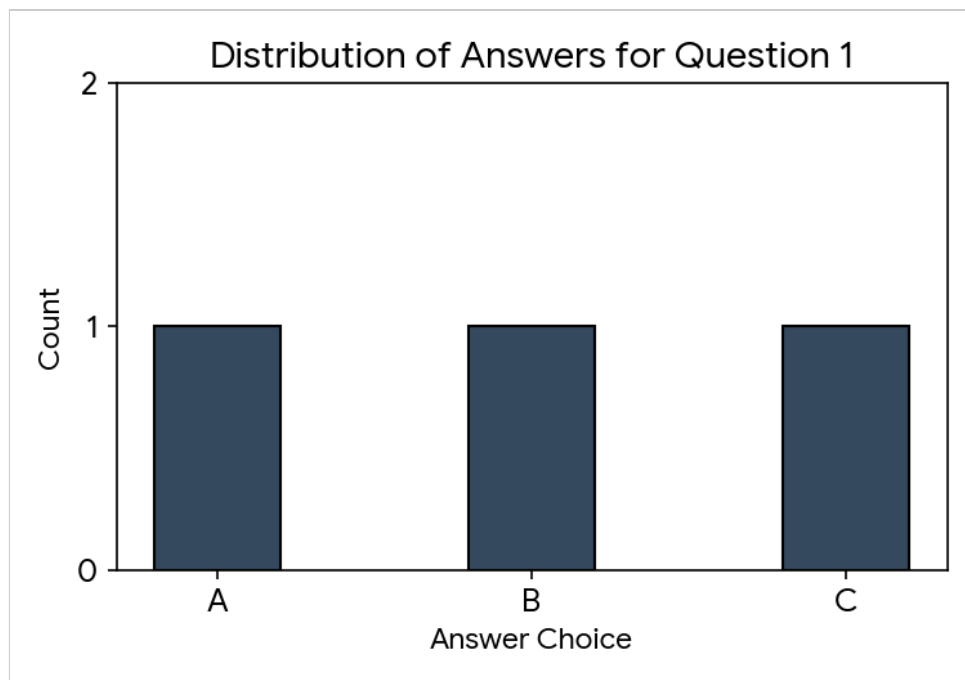
SET 10: SURVEY RESPONSES ANALYSIS

1. Create a grouped bar chart to visualize the distribution of answers for Question 1

R Code:

```
survey_data <- data.frame(Survey_ID=1:3, Question_1=c('A','B','C'),  
Question_2=c('B','A','A'), Question_3=c('C','D','B'))  
ggplot(survey_data, aes(x=Question_1)) + geom_bar(fill='#34495e',  
color='black', width=0.4) + labs(title='Distribution of Answers for Question  
1', x='Answer Choice', y='Count') + theme_minimal()
```

Output Chart:



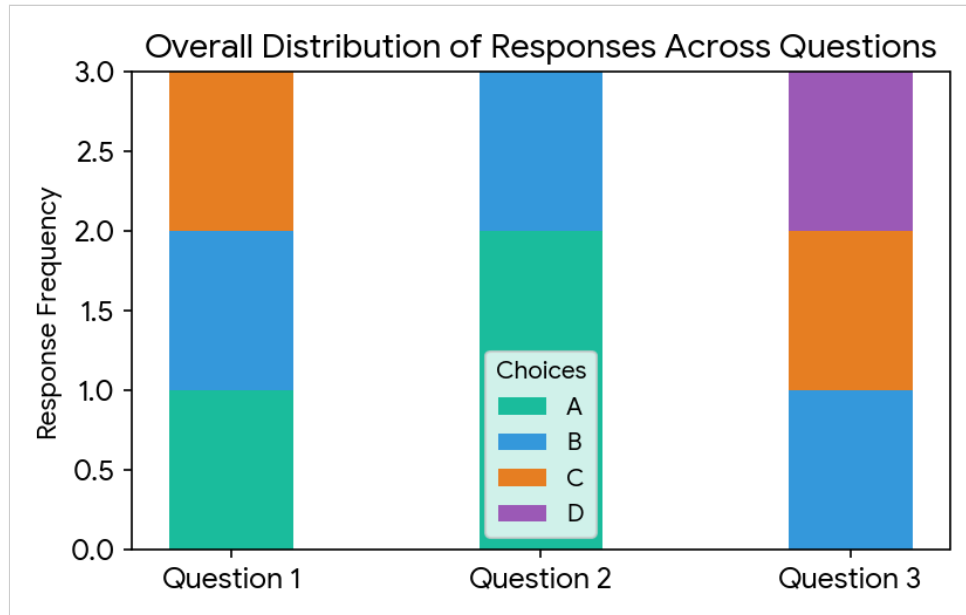
2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions

R Code:

```
melted_survey <- data.frame(Question=rep(c('Question 1','Question  
2','Question 3'), each=3), Response=c('A','B','C','B','A','A','C','D','B'))  
ggplot(melted_survey, aes(x=Question, fill=Response)) +  
geom_bar(position='stack', width=0.4) + labs(title='Overall Distribution of
```

```
Responses Across Questions', x='Question', y='Response Frequency') +  
theme_minimal()
```

Output Chart:



3. Build a table to show the survey response data for each survey

R Code:

```
print(survey_data)
```

Console Output View:

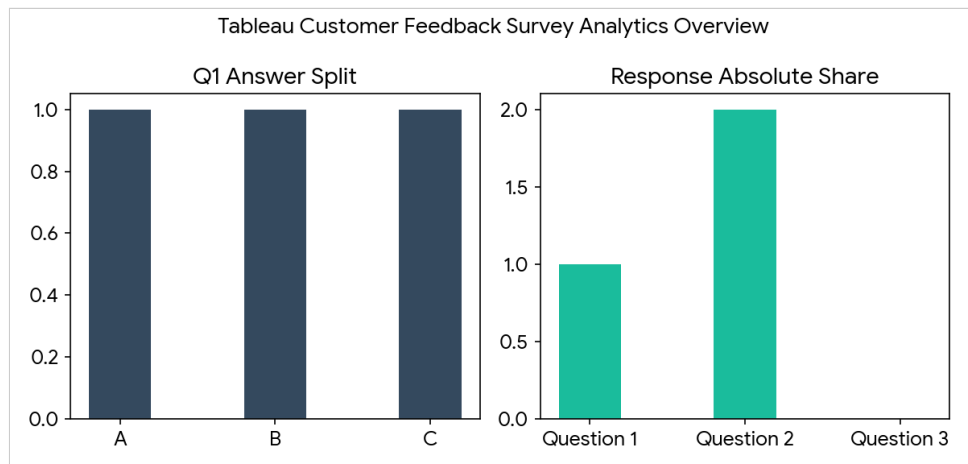
Survey Responses Raw Data Matrix			
Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

4. Develop a Tableau dashboard combining grouped bar chart, stacked bar chart, and the table

Tableau Steps:

1. Link 'set10_survey_responses.csv' into Tableau.
2. Sheet 1: Columns = Question 1, Rows = CNT(Survey ID) for the single bar split.
3. Sheet 2: Reshape fields to show cumulative responses, stacking columns via Mark card stack configurations.
4. Sheet 3: Drag columns to form the database grid matrix layout.
5. Assemble sheets onto the main interactive panel canvas dashboard workspace.

Dashboard Mock-up:



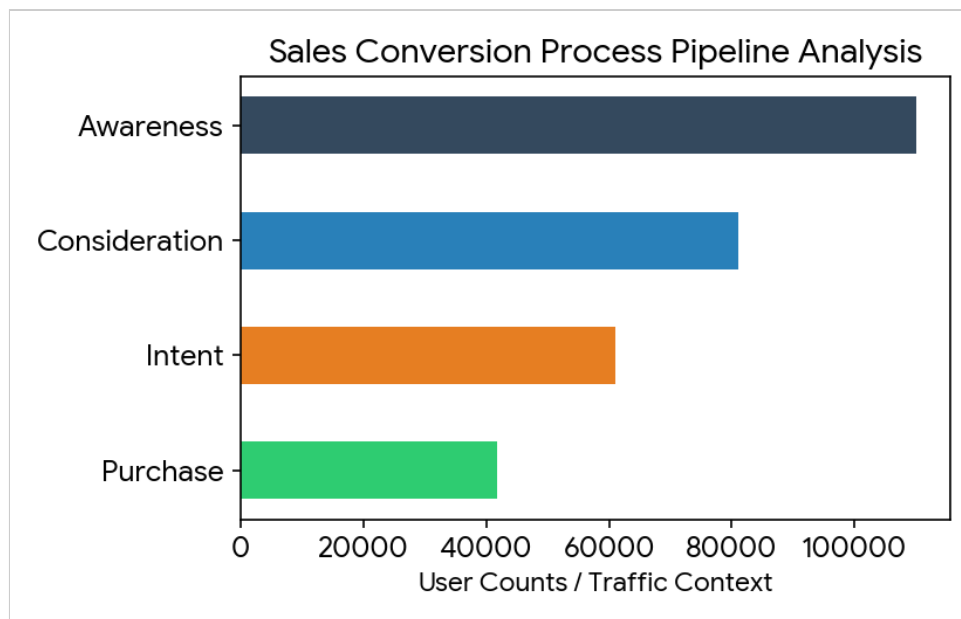
SET 11: PRODUCT CATEGORY ANALYSIS

1. Generate a funnel chart to analyze the sales conversion process for each product category

R Code:

```
funnel_data <-  
data.frame(Stage=factor(c('Awareness','Consideration','Intent','Purchase'),  
levels=c('Awareness','Consideration','Intent','Purchase')),  
Values=c(110000,81000,61000,41666))  
ggplot(funnel_data, aes(x=Stage, y=Values, fill=Stage)) +  
geom_bar(stat='identity', width=0.5, color='black') + coord_flip() +  
labs(title='Sales Conversion Process Pipeline Analysis', x='Stage', y='User  
Counts / Traffic Context') + theme_minimal()
```

Output Chart:



2. Build a table to display the sales data for each product category

R Code:

```
categories_data <-  
data.frame(Category=c('Electronics','Clothing','Appliances'),  
Sales=c(50000,35000,40000))  
print(categories_data)
```

Console Output View:

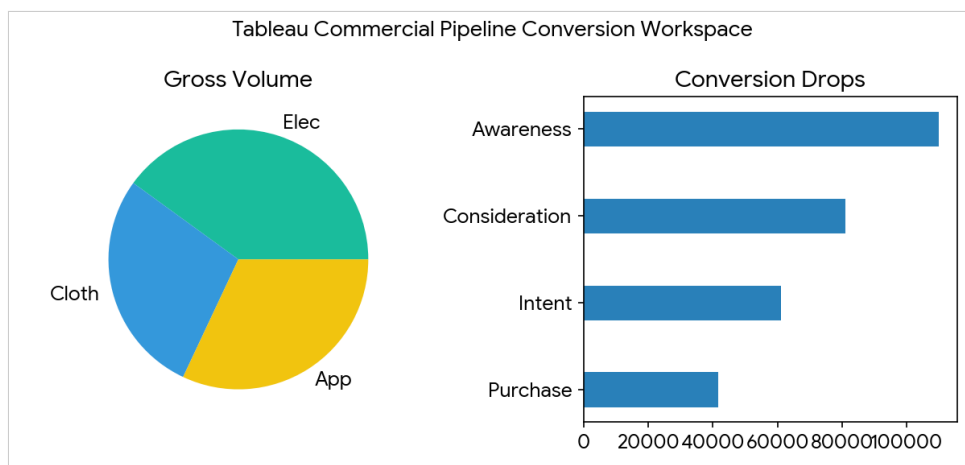
Product Categories Financial Volume Summary	
Category	Sales (in \$)
Electronics	50,000
Clothing	35,000
Appliances	40,000

3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table

Tableau Steps:

1. Load dataset 'set11_product_categories.csv' and 'set11_sales_conversion_funnel.csv'.
2. Create a pie layout chart split by category revenue parameters.
3. Build a funnel chart using descending horizontal count marks or a stepped bar shape layout matrix configuration.
4. Group components inside an interaction-driven live executive portal container.

Dashboard Mock-up:



4. Create a pie chart to represent the distribution of sales across product categories

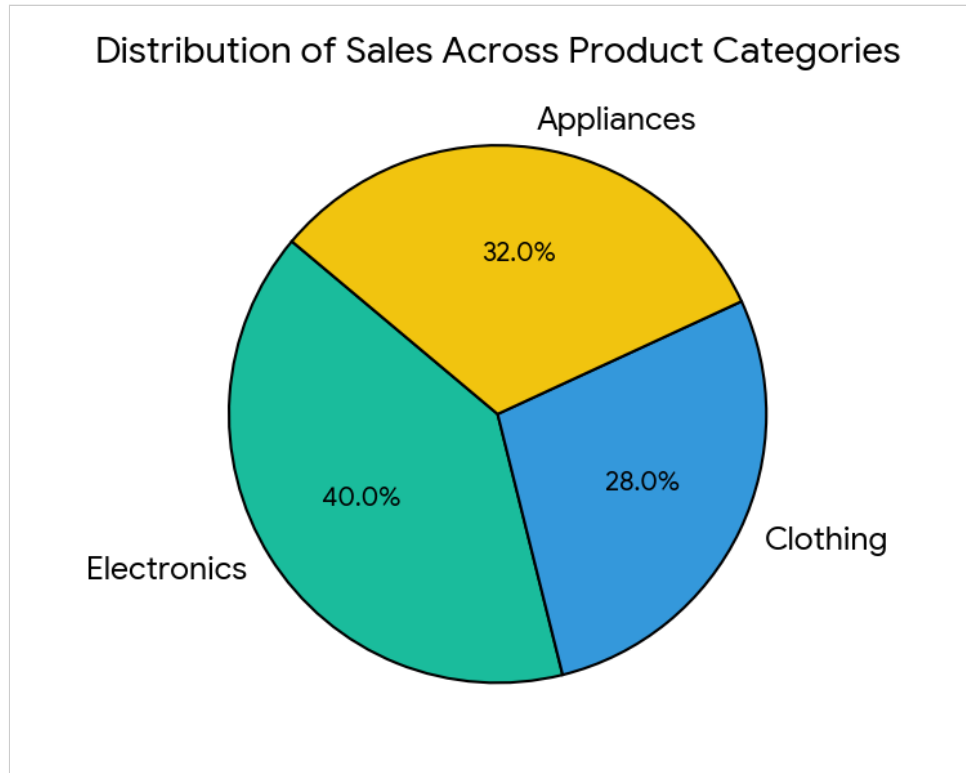
R Code:

```
ggplot(categories_data, aes(x='', y=Sales, fill=Category)) +  
geom_bar(stat='identity', width=1, color='black') + coord_polar('y', start=0)
```



```
+ labs(title='Distribution of Sales Across Product Categories') +  
theme_void()
```

Output Chart:



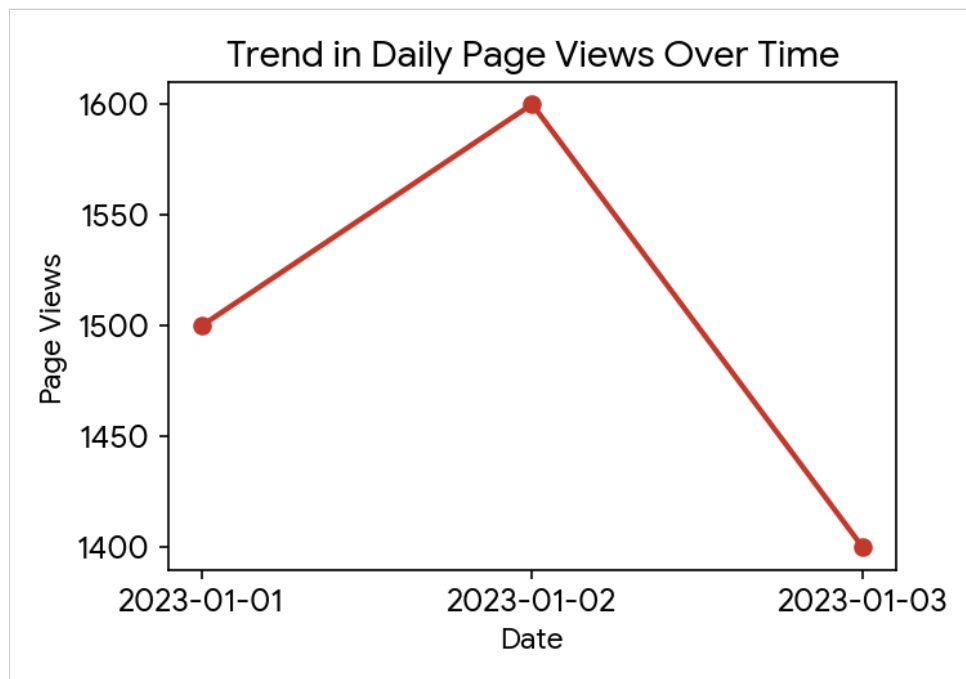
SET 12: WEBSITE TRAFFIC ANALYTICS

1. Create a line chart to visualize the trend in daily page views over time

R Code:

```
traffic_data <- data.frame(Date=c('2023-01-01', '2023-01-02', '2023-01-03'),  
Page_Views=c(1500,1600,1400), CTR=c(2.3,2.7,2.0))  
ggplot(traffic_data, aes(x=Date, y=Page_Views, group=1)) +  
geom_line(color='#c0392b', size=1.2) + geom_point(size=2.5) +  
labs(title='Trend in Daily Page Views Over Time', x='Date', y='Page Views') +  
theme_minimal()
```

Output Chart:

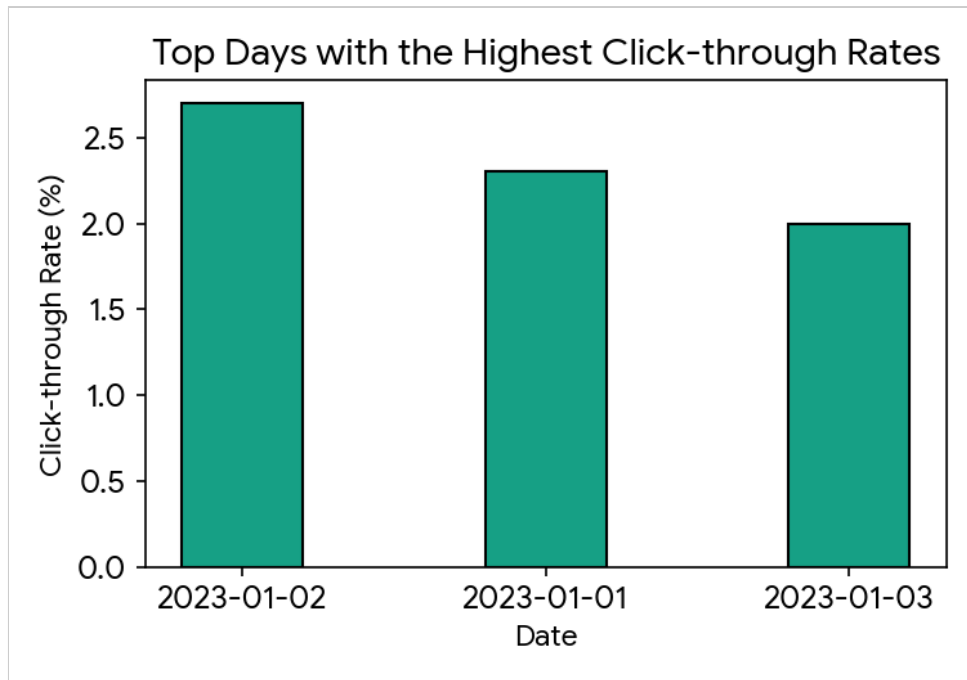


2. Generate a bar chart showing the top N days with the highest click-through rates

R Code:

```
ggplot(traffic_data, aes(x=reorder(Date, -CTR), y=CTR)) +  
geom_bar(stat='identity', fill='#16a085', color='black', width=0.4) +  
labs(title='Top Days with the Highest Click-through Rates', x='Date',  
y='Click-through Rate (%)') + theme_minimal()
```

Output Chart:



3. Build a table to show the website traffic data for each day

R Code:

```
print(traffic_data)
```

Console Output View:

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table

Tableau Steps:

1. Connect to data source file 'set12_website_traffic.csv'.
2. Sheet 1: Drag Date into Columns and Page Views into Rows for the baseline trend curve line view.
3. Sheet 2: Drag Date into Columns, CTR into Rows, and sort descending to isolate rankings cleanly.

4. Sheet 3: Generate a text list sheet to print fields as a grid data table view matrix element.
5. Combine all components inside a dynamic website tracking metric exploration board canvas.

Dashboard Mock-up:

